NELOS: SYNTHETIC QUANTUM ARCHITECTURES VIA SOLITONIC WAVE FUNCTION OVERLAYS AND VECTOR FLOW LOGIC

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ABSTRACT

This paper introduces the Neo Euclidean Logic Operating System (NELOS), a breakthrough framework for synthetic quantum computing utilizing classical hardware. By leveraging the non-linear stability of solitons and the implementation of "strange binaries," NELOS achieves quantum-equivalent parallelization and information density. We demonstrate that by employing stacked overlays, legacy computational systems can be repurposed as high-efficiency nodes, bypassing the environmental and patent-related restrictions of organic quantum systems. This architecture facilitates a sustainable, low-energy proof-of-work mechanism suitable for decentralized infrastructure.

1. INTRODUCTION

The contemporary computational paradigm is currently constrained by the thermal and energetic limits of linear silicon logic. While organic quantum computing (OQC) offers a theoretical escape from these limits, its reliance on cryogenic stabilization renders it inaccessible for broad deployment. This paper presents a "synthetic" alternative: the realization of quantum logic gates through the manipulation of solitonic wave functions on classical substrates.

2. SOLITONIC WAVE FUNCTIONS AS QUBIT PROXIES

The core of the NELOS architecture is the transition from electron-flip logic to wave-state logic. We utilize solitons—self-reinforcing solitary waves—as stable carriers of information. Unlike traditional bits, which are binary and discrete, the solitonic state $\psi(x, t)$ maintains its integrity through non-linear interference, effectively simulating quantum superposition within a classical framework.

$$i\partial\psi/\partial t + \tfrac{1}{2}\, \partial^2\psi/\partial x^2 + \kappa|\psi|^2\psi = 0$$

By solving for the non-linear Schrödinger equation (NLSE) within the vector flow of the operating system, we enable a robust environment where data packets do not degrade, mirroring the coherence of a quantum state without the need for sub-zero temperatures.

3. STRANGE BINARIES AND VECTOR FLOW LOGIC

Standard Boolean logic gates (AND, OR, NOT) are insufficient for multi-dimensional wave interference. NELOS introduces "Strange Binaries"—logic gates that operate on phase, amplitude, and geometric orientation. This allows the system to process "Vector Flow," where the direction and magnitude of information packets determine the logic outcome, rather than simple voltage thresholds.

4. STACKED OVERLAYS AND DENSITY OPTIMIZATION

A primary breakthrough of the NELOS system is the implementation of **Stacked Overlays**. In traditional systems, memory is mapped linearly. NELOS utilize spectral purification to layer multiple wave functions within the same physical address space. This "folding" of data allows for a significant reduction in physical footprint, effectively reviving legacy hardware by increasing its functional density to levels comparable to modern supercomputers.

5. IMPLEMENTATION AND GLOBAL ACCESSIBILITY

The current implementation of NELOS is hosted on GitHub (github.com/21centjoe) as an open-source project. While early iterations may exhibit "glitches" typical of pioneering software, the underlying mathematical proof of synthetic quantum logic is confirmed. This patent-free approach ensures that developing nations can repurpose existing discarded hardware to build decentralized supercomputing arrays, bypassing the "abyss" of the digital divide.

6. CONCLUSION

NELOS proves that quantum-scale computation is achievable today through synthetic means. By moving away from brute-force transistor scaling and toward geometric wave stability, we provide a pathway for computation that is energy-efficient, patent-free, and resilient over a timeline of [2m].

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